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Chapter 22

Configuring IP Services

Previous chapters have reviewed various dynamic routing protocols and how to perform basic switch and router configuration. This chapter delves deeper into options available for maintaining network routing connections, as well as how to work with some of the monitoring and diagnostic tools that are built into the Cisco IOS.

The Problems You'll Work On

In this chapter, you'll review questions concerning the following topics:

- Reviewing various High Availability (HA) options for reliable routing
- Managing and using Syslog servers and information
- Understanding SNMP v2 and v3
- Accessing performance and usage statistics using NetFlow

What to Watch Out For

Don't let common mistakes trip you up; watch for the following when working with these questions:

- Cisco does not provide its own Syslog server, so questions related to Syslog will focus on device configuration rather than server configuration. Many free and commercial servers are available to you, such as SolarWind's Kiwi Syslog Server and the open source syslog-ng.
- Both HA and dynamic routing protocols allow for recovery when routers become unavailable, but all HA solutions allow for recovery times in seconds rather than minutes. Both protocols can play a role in an effective enterprise network.
- Know the differences between SNMP v2 and v3, the latter of which is primarily support for authentication and encryption, and is more complex in setup.
- Understand the concepts of the NetFlow process, and be aware that the data sent to the NetFlow collectors includes statistics about all data that passes the router, and who the endpoints were on either side of the conversation. So you will see information such as "IP x.x.x.x had 5 conversations with IP y.y.y.y, and each conversation took x bytes and x packets to conclude." This is valuable information when attempting to diagnose network performance issues.
- Be aware that the server that collects, gathers, and analyzes NetFlow data is called a collector, and although Cisco has products that will perform the NetFlow collector function, many third-party products do so as well.

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Reviewing Various High Availability (HA) Options

890–906 Choose the best answer(s).

890. What configuration change is needed at the desktop workstation when activating a second router that has been configured using VRRP?

(A) A secondary router must be defined.

(B) The default gateway needs to be updated.

(C) No changes are required if the timer is set to 1 or above.

(D) No changes are required.

891. Which of the following are Cisco HA options for routing? (Select two.)

(A) DHCP

(B) GLBP

(C) RIP

(D) VRRP

892. What are requirements for VRRP to be used on a network? (Select two.)

(A) One router

(B) Two routers

(C) Additional virtual IP address

(D) Additional virtual MAC address

893. When using GLBP, if the AVG router fails, what router becomes the next AVG? (Select two.)

(A) The router with the highest IP address

(B) The first router that was configured for GLBP

(C) The router with the highest priority

(D) The router with the lowest IP address

894. What is the advertisement timer for VRRP?

(A) 1 second

(B) 2 seconds

(C) 5 seconds

(D) 10 seconds

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895. What is the major difference between GLBP and either VRRP or HSRP?

(A) VRRP supports faster failovers.

(B) GLBP is designed as an Active/Passive solution.

(C) GLBP performs Active/Active load balancing.

(D) HSRP supports faster failovers.

896. What statements are true about GLBP? (Select two.)

(A) Routers share a common virtual IP address.

(B) Routers have unique virtual IP addresses.

(C) Routers share a common virtual MAC address.

(D) Routers have unique virtual MAC addresses.

897. Which HA option for routers will have the lowest level of impact to users on the network during a failover?

(A) VRRP

(B) GLBP

(C) SSO

(D) HSRP

898. When working with GLBP, what is the role of the AVG?

(A) Is the active router processing all data

(B) Is the standby router waiting to assume the active role

(C) Manages which data will be processed by which router

(D) Send status messages to network management and syslog servers

899. What is the benefit offered by the SSO addition to HSRP?

(A) Preserves forwarding path

(B) Enables load balancing

(C) Enables a redundant path

(D) Sets the failover order

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900. When using GLBP, two routers are configured with the commands shown in the figure below.

```
Router1(config)#interface fastethernet 0/0
Router1(config-if)#ip address 10.1.8.11 255.255.255.0
Router1(config-if)#glbp 10 ip 10.1.8.10

Router2(config)#interface fastethernet 0/0
Router2(config-if)#ip address 10.1.8.12 255.255.255.0
Router2(config-if)#glbp 10 ip 10.1.8.10
```

The AVG role is running on Router1 with the highest priority. Router1 is rebooted and the AVG role is transferred to Router2. When will the AVG role move back to Router1?

(A) When Router1 comes back on line

(B) 5 minutes after Router1 comes back on line

(C) Never.

(D) When Router2 is loses network connectivity

901. You are configuring VRRP and have used the commands shown in the figure below. It is not working. What changes would you make to the configuration to make it work?

```
Router1# configure terminal
Router1(config)# interface GigabitEthernet 0/0/0
Router1(config-if)# ip address 172.16.6.5 255.255.255.0
Router1(config-if)# vrrp 10 ip 172.16.6.1
Router1(config-if)# end

Router2# configure terminal
Router2(config)# interface GigabitEthernet 0/0/0
Router2(config-if)# ip address 172.16.6.6 255.255.255.0
Router2(config-if)# vrrp 10 ip 172.16.6.1 secondary
Router2(config-if)# end
```

(A) Change the interface IP address to match on both routers.

(B) Issue shutdown and no shutdown on the interfaces.

(C) VRRP may not be supported on these routers.

(D) Three routers are required for VRRP.

(E) All routers require matching VRRP primary and secondary IP addresses.

902. Which of the following are valid GLBP debug commands? (Select two.)

(A) `debug glbp`

(B) `debug glbp errors`

(C) `debug glbp data`

(D) `debug glbp packets`

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903. You have configured HSRP but want to add SSO. What changes do you need to make to Router2 to complete this configuration?

(A) Use the following commands:

Router1(config)#`redundancy`

Router1(config)#`mode sso`

Router1(config)#`exit`

(B) Use the following commands:

Router1(config)#`redundancy`

Router1(config)#`mode sso`

Router1(config)#`exit`

Router1(config)#`no standby sso`

Router1(config)#`standby sso`

(C) Use the following commands:

Router1(config)#`mode sso`

Router1(config)#`no standby sso`

Router1(config)#`standby sso`

(D) Use the following commands:

Router1(config)#`redundancy`

Router1(config-red)#`mode sso`

Router1(config-red)#`exit`

Router1(config)#`no standby sso`

Router1(config)#`standby sso`

904. Members of a GLBP group communicate with hello messages using the IP address of _____ on ____ port _____.

(A) 224.0.0.100, TCP, 3222

(B) 225.0.0.55, UDP, 3222

(C) 224.0.0.102, UDP, 3222

(D) 225.0.0.55, TCP, 3222

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905. The output shown in the figure below is from the show glbp command. You are having problems creating the router group to support HA routing. What options should you examine on the two devices? (Select two.)

```
Router# show glbp 10

FastEthernet0/0 - Group 10
  State is Active
    2 state changes, last state change 23:50:33
  Virtual IP address is 10.21.8.10
  Hello time 5 sec, hold time 18 sec
    Next hello sent in 4.300 secs
  Redirect time 1800 sec, forwarder time-out 28800 sec
  Authentication text "stringabc"
  Preemption enabled, min delay 60 sec
  Active is local
  Standby is unknown
  Priority 254 (configured)
  Weighting 105 (configured 110), thresholds: lower 95, upper 105
    Track object 2 state Down decrement 5
  Load balancing: host-dependent
  There is 1 forwarder (1 active)
  Forwarder 1
    State is Active
      1 state change, last state change 23:50:15
    MAC address is 0007.b400.0101 (default)
    Owner ID is 0005.0050.6c08
    Redirection enabled
    Preemption enabled, min delay 60 sec
    Active is local, weighting 105
```

(A) Group numbers match

(B) Priorities match

(C) Preemption settings match

(D) Hello times match

(E) Virtual IP addresses match

906. How many standby groups can be assigned to a router when using HSRP?

(A) 1

(B) 16

(C) 256

(D) 512

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Managing and Using Syslog Servers

907–914 Choose the best answer(s).

907. How does a syslog server separate incoming log information?

(A) By client MAC address

(B) By client IP address

(C) It does not separate information.

(D) It depends on the server.

908. What command will display the settings of logging on your switch or router?

(A) `display logging`

(B) `show logging`

(C) `show log options`

(D) `show log settings`

909. When logging messages to a syslog server, what severity levels are you able to send to the syslog server?

(A) Only level 0

(B) Below level 2

(C) Below level 5

(D) All severity levels

910. You have received the information shown in the figure below on your syslog server. What does this information tell you about your devices on the network?

```
*Mar  6 22:48:34.452 UTC: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0,
    changed state to up
```

(A) There was a change in the state of the gigabit 0/0 interface.

(B) Something happened March 5.

(C) There was a message sent by the device kernel.

(D) There was a change in the state of the loopback0 interface.

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911. You have configured your router using the commands shown in the figure below, but your Syslog server is not showing any data. Why not?

```
Router1#configure terminal
Router1(config)#service timestamps log datetime
Router1(config)#logging host 10.1.8.50
Router1(config)#logging trap warnings
```

(A) Your traffic is being blocked by NAT translation.

(B) You need to copy the running-configuration startup-configuration.

(C) The syslog server is configured for a non-standard port.

(D) The router needs to be restarted for the configuration to take effect.

912. What is the default port on which a syslog server operates?

(A) UDP 110

(B) UDP 514

(C) TCP 514

(D) TCP 110

913. What elements make up a basic syslog message? (Select two.)

(A) Time

(B) Link speed

(C) Severity

(D) IOS Version

914. You have been having issues with your router, which becomes unresponsive and will not respond to any of the administrative interfaces. Your only option is to reboot the router with a power reset. What tool can help you to identify the issue that is affecting the router?

(A) SNMP

(B) TFTP

(C) Syslog

(D) SMTP

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Understanding SNMP v2 and v3

915–924 Choose the best answer(s).

915. What is the name given to an alert generated by an SNMP agent?

(A) Capture
(B) Log
(C) Syslog
(D) Trap

916. Which of these are examples of SNMP Trap receivers? (Select two.)

(A) HP Openview
(B) Oracle SQL
(C) Nagios
(D) Java

917. What is the name given to a piece of data that can be retrieved via SNMP?

(A) MIB
(B) DATA
(C) RFC
(D) OID

918. What is the default port used by SNMP?

(A) UDP 125
(B) TCP 161
(C) TCP 125
(D) UDP 161

919. What is name of the documented list of OID for a product?

(A) RFC
(B) MIB
(C) IETF
(D) SSH

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920. What is the purpose of a community name in SNMP?

(A) Host selection

(B) Security grouping

(C) OID selection

(D) Security password

921. What is the major advantage of SNMP v3 over SNMP v2? (Select two.)

(A) User authentication

(B) 64-bit community names

(C) Bi-directional handshaking

(D) Data encryption

(E) Customizable ports

922. What are the two standard permission types used by SNMP? (Select two.)

(A) Query

(B) Read

(C) Write

(D) Set

923. What security step can be used to secure SNMP?

(A) Enable NAT on the device

(B) Implement case-sensitive community names

(C) Implement access lists

(D) RSA-encode community names

924. What are options for securing SNMP traffic on your network? (Select three.)

(A) Using secure community names

(B) Restricting access to read-only

(C) Using IOS version 12.5.1

(D) Implementing access lists

(E) Enabling password encryption

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Accessing Statistics Using NetFlow

925–927 Choose the best answer(s).

925. What information is required to send NetFlow data to a collector? (Select two.)

(A) The IP address of the collector

(B) The DNS hostname of the client

(C) The port the collector is using

(D) The NetFlow proxy address

926. Which of the following are components in NetFlow data analysis? (Select two.)

(A) Firewall

(B) Syslog server

(C) NetFlow-enabled device

(D) NetFlow collector

927. You have configured NetFlow to send data to your collector using the four lines shown in the figure below.

```
Router1(config)#interface gigabit 0/0
Router1(config-if)#ip route-cache flow
Router1(config-if)#exit
Router1(config)#ip flow-export 10.1.1.25
```

The collector is not showing any data. What is the likely cause of the problem? (Select two.)

(A) The collector is configured for a non-default port.

(B) The router does not support sending NetFlow data.

(C) The collector is using a different version.

(D) The router must be using a dynamic routing protocol.